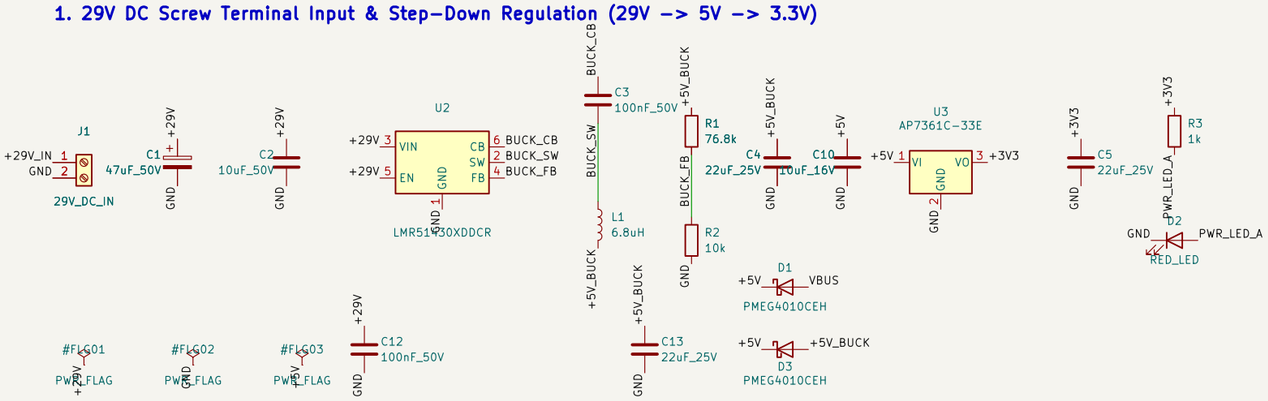
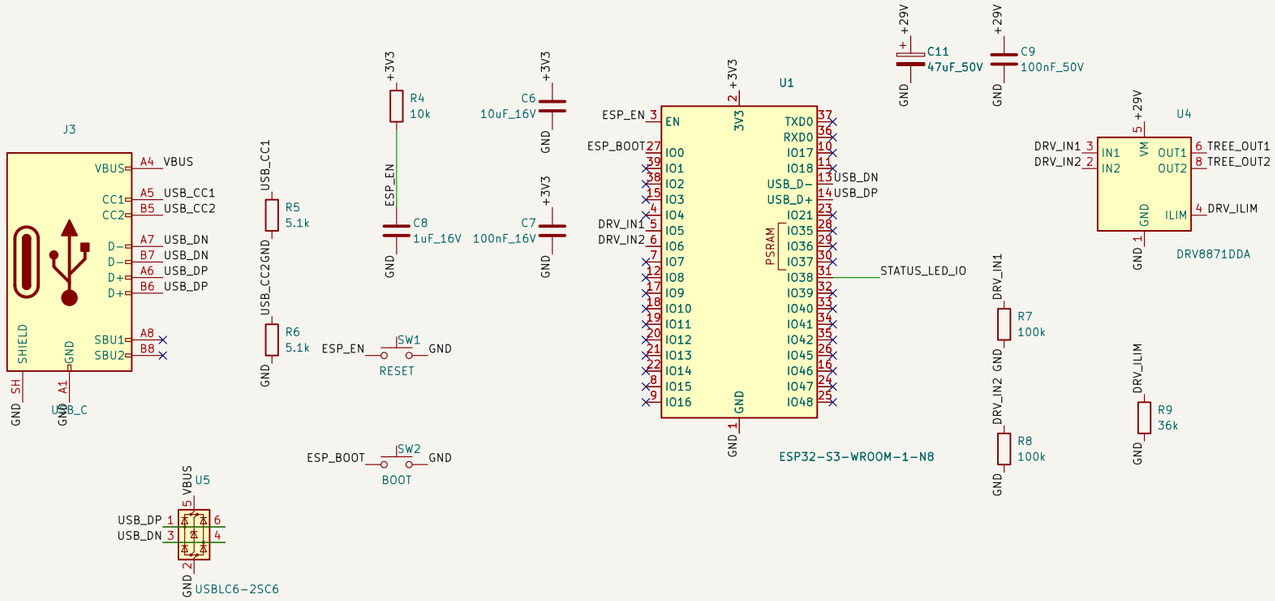
1. 29V DC Screw Terminal Input & Step-Down Regulation (29V -> 5V -> 3.3V)



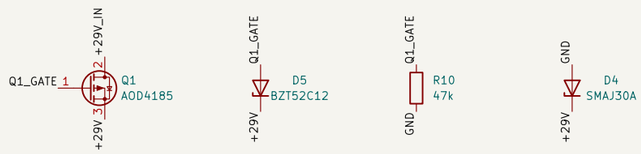
4. DRV8871 2A H-Bridge, Screw Terminal (J2) & Plated Tree Pads

2. USB-C Native Serial/JTAG & Reset/Boot Controls

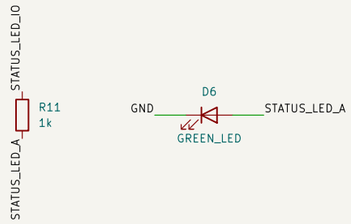
3. ESP32-S3-WROOM-1-N8 Matter Controller



5. 29V Input Reverse-Polarity P-MOSFET & TVS Clamp



6. Matter Status LED (GPIO38)



7. Diagnostic Rail Probe Pads



Matter over Wi-Fi (SmartThings Single-Card Extended Color Light)
29V 2A H-Bridge – Reverse-Polarity P-MOSFET – USB-C ESD & Low-Leakage Schottky OR

Pixel Paw Studio

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File: christmas_tree_controller.kicad_sch

Title: ESP32-S3 Matter Christmas Tree Controller

Size: A3	Date: 2026-11-10	**Rev: C**
KiCad E.D.A. 10.0.8		Id: 1/1